

Free Cloud & Container Labs

Zero-Cost Hands-On Practice — Cloud Admin & Containers

Bottom line: You can get genuinely job-ready hands-on experience without spending a dollar on lab platforms. Between AWS Educate, Microsoft Learn sandboxes, Killercode, LocalStack, and Docker Desktop — you have a complete free lab environment. This guide covers exactly what's free, what's not, and what to use first.

Heads up — Play with Kubernetes is gone. It was shut down March 1, 2026. Killercode is the replacement. Everything else in this guide is current as of June 2026.

1 — Free Cloud Labs: AWS & Azure

These let you practice in real or near-real cloud environments without a bill. Some use your own free-tier account, some use sandboxed accounts so you don't need a credit card at all. Both options are listed.

No credit card at all? Use Microsoft Learn Sandbox and Killercode — both give you real environments with zero payment info required. AWS and GCP do ask for a card even on free tier (though they won't charge if you stay within limits).

AWS Educate — Free Labs (No Credit Card)

FREE. AWS's student/learner platform. Register with just an email — no credit card. Gives you access to free hands-on labs in real AWS environments. Shareable digital badges on completion. Ideal starting point.

AWS Skill Builder — Free Tier

FREE. 1,000+ free AWS courses plus free Builder Labs (10 free labs included). The SysOps Administrator learning path maps directly to SOA-C02. No credit card needed for courses — only the live sandbox labs require one.

AWS Free Tier — 12 Months

FREE (requires credit card, won't charge if you stay in limits). Real AWS console for 12 months. EC2, S3, VPC, IAM, CloudWatch — the actual services on SOA-C02. Set a \$5 billing alert immediately after signup.

Microsoft Learn — Azure Sandbox (No Credit Card)

FREE. No credit card, no Azure account needed. Microsoft Learn's guided exercises spin up a real sandboxed Azure environment. Full AZ-104 learning path is here. Best no-cost Azure lab option available.

Azure Free Account — 12 Months

FREE for 12 months (requires credit card, won't charge within limits). Real Azure portal. Practice everything on AZ-104 exam. Same billing alert advice — set one at \$5.

labTpro — Free Azure Labs (12 Free, No Card)

FREE. 12 guided Azure lab simulations with no subscription or Azure account required. Uses high-fidelity portal simulations. Covers VMs, storage, networking, users. Full 85+ lab access is \$4.99 one-time — not a subscription.

Google Cloud Skills Boost — Free Paths

FREE learning paths and some free labs. Even if you're AWS/Azure focused, GCP labs reinforce cloud concepts that transfer. No credit card for the learning paths.

LocalStack — Free Non-Commercial Tier

FREE for non-commercial/student use. Runs AWS services locally on your machine inside Docker — S3, Lambda, DynamoDB, SQS and more — without touching real AWS. Note: changed pricing in March 2026, now requires a free account. Students get free access via GitHub verification.

AWS Well-Architected Labs (Free)

FREE. AWS's official hands-on labs organized around the five pillars: Security, Reliability, Performance, Cost, Operations. Real tasks in your own AWS account. Excellent for SOA-C02 prep.

freeCodeCamp — AWS Full Course (YouTube)

FREE. Full-length AWS Cloud Practitioner and Solutions Architect courses on YouTube. Search 'AWS' on the freeCodeCamp channel. No account needed, watch anytime.

2 — Free Container Labs: Docker & Kubernetes

Containers are a hands-on skill — you can't learn them by reading. All of the below are free. Play with Kubernetes (Docker's old playground) was shut down March 1, 2026 — Killercoda is the replacement.

Killercoda — Free Browser Labs ★ START HERE

FREE. The best free container lab platform after Play with K8s shut down. Browser-based Linux, Docker, and Kubernetes environments — no install, no account needed for most labs. Has dedicated CKA prep scenarios. Use this daily.

Play with Docker — Free Sandbox

FREE. Docker's official browser sandbox. Real Linux environment with Docker pre-installed. 4-hour sessions, then resets. Practice container commands, Dockerfiles, and compose without touching your machine.

Docker Desktop — Free (Personal Use)

FREE for personal/educational use. Installs Docker on your Windows or Mac laptop. Includes a local single-node Kubernetes cluster you can enable with one click. Your primary local container lab environment.

KodeKloud Studio — Free Docker/K8s Labs

FREE browser-based Docker and Kubernetes playgrounds from KodeKloud. No account required. Good for quick practice sessions without setting up anything locally.

LabEx — Free Kubernetes Labs

FREE. Browser-based Kubernetes labs with guided exercises. Covers cluster management, deployments, services, and scaling. Good supplement to Killercoda.

Minikube — Free Local K8s

FREE. Runs a single-node Kubernetes cluster on your laptop. The standard tool for local K8s practice. Install alongside Docker Desktop. Used in most Kubernetes tutorials you'll find.

Kind (Kubernetes in Docker) — Free

FREE. Runs multi-node Kubernetes clusters inside Docker containers on your machine. Lighter than Minikube, good for simulating multi-node setups. Used heavily in CKA prep.

TechWorld with Nana — Docker & K8s (YouTube)

FREE. Best free video content for Docker and Kubernetes. Watch 'Docker Tutorial for Beginners' then 'Kubernetes Tutorial for Beginners' — both on her channel, both free, both excellent.

freeCodeCamp — Docker Full Course (YouTube)

FREE. Full-length Docker and Kubernetes courses on YouTube. Search 'Docker' or 'Kubernetes' on the freeCodeCamp channel. No account needed.

GitHub — Free DevOps Home Lab Guide

FREE. freeCodeCamp guide to building a full local DevOps lab with Docker, Kubernetes, Prometheus/Grafana, and Ansible — all on your own machine. Zero cloud cost. Covers a realistic end-to-end setup.

3 — Free Cert Study: SOA-C02, AZ-104 & More

You still need to pay for the exam itself — but everything leading up to it can be free. These are the best zero-cost study resources for your next certs.

AWS Skill Builder — SysOps Path (Free)

FREE. Official Amazon SOA-C02 learning path. Courses, videos, and practice questions mapped to every exam domain. Your primary free SOA-C02 study resource.

Microsoft Learn — AZ-104 Path (Free)

FREE. Official Microsoft AZ-104 learning path with sandbox labs included. Covers all exam domains. No Azure account or credit card needed for the guided exercises.

TutorialsDojo — Free Cheat Sheets

FREE. Comprehensive AWS and Azure service cheat sheets. One page per service — what it does, key features, exam gotchas. Invaluable for last-week exam prep.

Professor Messer — Free CompTIA Videos

FREE. Full video courses for A+, Network+, and Security+ on YouTube and his site. Already relevant to you for any CompTIA renewals or upgrades like CASP+.

HashiCorp — Free Terraform Tutorials

FREE. Official Terraform learning path from HashiCorp. Covers everything on the Terraform Associate exam. Start with 'Get Started — AWS' or Azure depending on your focus.

Linux Foundation — Free CKA Prep

FREE. Free Kubernetes and Linux courses from the org that runs the CKA exam. Vendor-neutral, high quality. Use alongside Killercode labs for CKA prep.

r/AWSCertifications (Reddit)

FREE. Active Reddit community sharing SOA-C02 exam experiences, study tips, and free resources. Read recent 'passed' posts to understand what the current exam focuses on.

r/AzureCertification (Reddit)

FREE. Same idea for AZ-104. Real test-takers sharing what was on the exam, which study materials worked, and passing strategies.

Recommended Free Learning Sequence

Cloud — Day 1: Register for AWS Educate (no credit card). Complete the free Cloud Operations course. This gives you structured SOA-C02 prep without spending anything.

Cloud — Week 1: Start the Microsoft Learn AZ-104 path. Use the built-in sandbox labs — no Azure account needed. Work through the VM, storage, and networking modules.

Cloud — Ongoing: Sign up for AWS Free Tier (requires card, no charges within limits). Set a \$5 billing alert immediately. Practice EC2, VPCs, IAM, and S3 in the real console.

Containers — Day 1: Install Docker Desktop (free). Run your first container: 'docker run hello-world'. Then work through the official Docker Getting Started tutorial.

Containers — Week 1: Go to Killercode.com. Work through the Docker and Kubernetes beginner labs in your browser. No install, no account. This is your daily container practice platform.

Containers — Week 2+: Enable Kubernetes in Docker Desktop (one checkbox). Deploy your first app to a local cluster using kubectl. Follow TechWorld with Nana's Kubernetes beginner playlist on YouTube.

Cert Prep — Final 2 Weeks: TutorialsDojo cheat sheets for the service summary. Read r/AWSCertifications or r/AzureCertification for recent exam reports. Then take the exam.

Total cost to get hands-on with cloud and containers: \$0. The only thing you pay for is the exam itself when you're ready. SOA-C02 is ~\$300. AZ-104 is ~\$165. Terraform Associate is ~\$70.